Wayne Williams Opanja

Software & AI Engineer · Builds production AI products end to end

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SELECTED IMPACT

- **Shipped national govtech:** led frontend and UX for Nigeria's Federal Ministry of Education RICC platform, now live at ricc.education.gov.ng, and worked on the federal Trademark Registry.
- **Put AI to work for a county government:** built a WhatsApp assistant that answers Nairobi City County revenue questions from 7 Excel report formats, with read-only, validated SQL so the model cannot change data.
- **Turned 79 guidance documents into working software:** built British International Investment's climate and nature risk toolkit for financial institutions: 23 modules and 20 guided workbenches.
- **Founded and run an AI company:** Pneuma Labs has delivered 8 projects for 8 clients across the UK, Kenya, Uganda, Rwanda, Burundi and the DRC.

SUMMARY

Software and AI engineer who takes products from the first discovery call to production. I build LLM agents, RAG and text-to-SQL systems, WhatsApp-first products and full-stack web apps in TypeScript and Python, and I put guardrails around AI so it can be trusted with real data. Background in data science, climate-risk analytics and security (CHFI). Fluent in English and Swahili.

CORE SKILLS

AI: LLM agents and multi-agent pipelines, RAG, text-to-SQL, tool calling, provider fallback, output validation, OCR and document parsing, scikit-learn, PyTorch.

Engineering: TypeScript, JavaScript, Python, Java, PHP; Next.js, React, Node.js, NestJS, Express, Django, FastAPI; PostgreSQL, Supabase, Prisma, MongoDB; REST APIs, Kafka.

Delivery: Docker, Kubernetes, GitHub Actions, Vercel, Render, AWS, Azure, GCP; Vitest, Jest, pytest, Playwright; WhatsApp, Twilio, Paystack, Smile ID, Sumsub.

Product: discovery and user research, requirements, UX and prototyping, QA and UAT, client and stakeholder management.

EXPERIENCE

Founder & Lead Product Engineer · Pneuma Labs | Nairobi, clients in 6 countries | 2026 – Present

AI and automation company: 8 projects for 8 clients in logistics, real estate, mining, financial services and government.

- **Nairobi City County revenue assistant:** wrote tested parsers for 7 Excel report layouts into PostgreSQL and an LLM agent that answers staff questions on web and WhatsApp. Ran accuracy audits against the source sheets that caught and fixed double-counting and date-range errors.
- **NSE market research system:** built a multi-agent pipeline (analysts, bull/bear debate, trader, risk check) that runs every morning before market open, scores stocks against an inflation-adjusted hurdle and logs every call so accuracy is measured. Falls back across 4 LLM providers; used by analysts in Kenya.
- **PropFlow property-management SaaS:** designed and built billing, arrears, leases with e-signing, maintenance and a WhatsApp tenant line with emergency voice calls. 90 data models, 63 screens, 120 test files, Paystack payments, nightly backups. AI suggests, code validates, then acts.
- **Process automation:** mapped operations for Mediscope and the Strathmore Centre for Value Investing and designed automations around how their teams work.

Consultant, Climate & Nature Toolkit · British International Investment | Remote | 2026 – Present

- **Made a development-finance framework usable:** built the platform behind BII's climate and nature risk toolkit for financial institutions: client climate screening, maturity assessment, financed-emissions and stress-test calculators, with PDF, Excel and Word exports (Next.js, TypeScript, Prisma, PostgreSQL).

Senior Software Engineer · ProDG Studios | Remote, UK-based | 2025 – 2026

- **RICC, Federal Ministry of Education, Nigeria:** led frontend and UI/UX. Researched why universities, industry and regulators weren't connecting and turned that into the user journeys for innovation assessment and routes to funding and IP support.
- **Trademark Registry, Federal Ministry of Industry, Trade and Investment, Nigeria:** built digital registration and administration workflows that replaced manual processes.
- **Venture Garden Group:** research, product design, development, QA and UAT on a platform for diaspora investment, including KYC integrations with Smile ID and Sumsub.
- **NGAO, Action Lab:** built a WhatsApp-first AI data-collection platform that turns conversational answers into structured records for reporting.

Data Scientist, then Partner · Climate Shapers Africa | Nairobi | 2025 – Present

- **Climate risk for banks:** built climate-stress exposure analytics for the Kenya and Tanzania Bankers Associations, linking climate indicators to probability of default, with NCBA, Equity Bank and Victoria Commercial Bank among the stakeholders.
- **Data products:** built the Africa Climate & Nature Data Platform prototype (10 sector dashboards, Kenya county risk map), Power BI dashboards and AI research automations for the Kenya Climate & Nature Directory. Partner since 2026.

Earlier experience

- **Strathmore Institute of Mathematical Sciences, Student Assistant (Developer) (2023 – 2025):** built a course evaluation system (Django, React, PostgreSQL) that raised student feedback rates by 75%.
- **Harambee SACCO, Penetration Testing Consultant (2024):** tested a financial cooperative's systems and wrote up findings for remediation.
- **Strathmore University ICT, Software Developer Intern (2025 – 2026, degree internship):** rebuilt the phone-billing system as an Odoo module with call-record and HR/ERP sync.

SELECTED PROJECTS

- **MessageHeard v3 (team project, 2025 – 2026):** worked on a multi-tenant AI video-generation platform with Clerk auth, role-based access, transcript-to-scene breakdown and a job queue for concurrent image and video generation.
- **Chat Your Finances (lead developer):** AI assistant that parses M-Pesa, bank and payslip PDFs and answers questions on WhatsApp, Telegram and web, with a KRA tax engine. 98 API routes, 31 test files.

EDUCATION AND CERTIFICATIONS

- **Strathmore University:** Bachelor of Business Information Technology (2024 – 2027, in progress, Grade A average); Diploma in Business Information Technology (2023 – 2024).
- **Certifications:** CHFI, EC-Council (2024); Applied Data Science & ML, WorldQuant University (2025); Google Cybersecurity; Simplilearn Full-Stack Development.
- **Languages:** English and Swahili (fluent), German (beginner).